

Code-Layout, Readability and Reusability

Date	29 September 2026
Team ID	SWTID-2026-1393
Project Name	EduGenie – AI Learning Assistant
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability: This document evaluates the EduGenie codebase in terms of its structure, readability and reusability. The assessment uses the checklist provided in the supplied template. The project is organized into separate Python modules and frontend folders, supporting modular development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Python and frontend code follow a consistent indentation structure.
2	Proper File Structure	Files and folders are logically organized	Yes	Frontend resources are separated into static/templates and backend modules are separated into Python files.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Module and function-related names such as explanation, learning path, Q&A, quiz and summary are descriptive.
4	Function / Method Names	Functions are descriptively named	Yes	Functional modules are named according to their learning-assistance purpose.
5	Code Comments	Inline and block comments explain logic	Partial	Comments can be used where complex AI/API processing needs additional explanation.
6	Modular Design	Code is split into reusable functions/modules	Yes	Learning features are separated into dedicated modules, supporting reuse and maintenance.
7	No Redundant Code	Duplicate or unused code is removed	Partial	The modular structure reduces duplication; a final cleanup review can further remove unused code.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Backend/API operations can be handled through application error handling and validation.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	Explanation Module	Python	AI-generated explanations and topic learning assistance	High
2	Learning Path Module	Python	Personalized learning-path generation	High
3	Q&A; Module	Python	Question-answering functionality	High
4	Quiz Module	Python	Quiz generation and assessment support	High
5	Summary Module	Python	Summary and study-note generation	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	4	Logical separation of frontend folders and Python learning modules supports maintainability.
Readability	4	Descriptive module names and a clear feature-based organization improve readability.
Reusability	4	The explanation, learning-path, Q&A, quiz and summary modules can be reused across learning requests.
Documentation / Comments	3	Project documentation is present; additional inline comments for complex processing would improve clarity.
Overall Score	4	Overall, the project demonstrates good structure, readable organization and reusable modular components.

Team Members – Group 1

No.	Member Name	Group No.
1	BALAMURUGAN M	1
2	Aakash A	1
3	Akash S	1
4	Santhosh kumar R	1
5	Gunaseelan K	1